

MATH 1203 CIS & BIT DISCRETE MATHEMATICS

Lecture 1: COUNTING TECHNIQUES

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Lesson Outcome

By the end of this lesson you should be able to:

- Define product rule as a counting techniques
- Demonstrate understanding and ability to solve problems using product rule technique.
- Describe how product rule can be applied in solving one - one functions
- Define Sum rule as a counting techniques
- Demonstrate understanding and ability to solve problems using sum rule technique.
- Combine Product rule and sum rule to solve more complex problems

INTRODUCTION

- I present to you the first lecture of this course on the topic, Counting techniques.
- In the previous lecture, you learnt about the ordered pairs, relation and binary composition
- In this lecture, you will be introduced to the basic techniques for counting.
- We will Look on the product rule and Sum rule as counting techniques
- how we can combine both sum and product rule to solve complex problems

The Product Rule

Definition 1

We define the product rule as procedure that can be broken down into a sequence of two tasks. If there are n_1 ways to do the first task and n_2 ways to do the second task then there are $n_1 \cdot n_2$ ways to do the procedure.

Now we will discuss Examples 2-7 which show how product rule is applied in solving in different problems.

Product Rule CONT'D

Example 2

Suppose a College has 3 different history courses, 4 different literature courses and 2 different sociology courses. How many ways can a student choose one of each kind of courses?

Solution: The number m of ways a student can choose one of each kind of courses is:

$$\begin{aligned} m &= n_1 \cdot n_2 \cdot n_3 \\ &= 3 \cdot 4 \cdot 2 \\ &= 24 \end{aligned}$$

Product Rule CONT'D

Example 3

A new company with just two employees Alice and Bob rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?

Solution

- The procedure for assigning offices to these two employees consists of assigning an office to Bob , which can be done in 12 ways.

Product Rule CONT'D

- Assigning an office to Alice different from the office assigned to Bob, which can be done in 11 ways.
- By product rule there are $12 \cdot 11 = 132$ ways to assign offices to these two employees.

Counting bit Strings

Example 4

How many bit strings of length seven are there?

Solution

- Each of the seven bits can be chosen in two ways , because each bit is either 0 or 1.
- Therefore the product rule shows there are a total of $2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^7 = 128$ different bit strings of length seven.

Counting License Plates

Example 5

How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits?

Solution

- There are 26 choices for each of the three letters i.e. $26 \cdot 26 \cdot 26$.
- And ten choices for each of the three digits i.e. $10 \cdot 10 \cdot 10$.
- Hence by product rule there are a total of $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 17,576,000$ different possible

Counting Functions

Example 6

How many functions are there from a set with m elements to a set with n elements?

Solution

- A function represents a choice of one of the n elements of the co-domain for each of the m elements in the domain.
- Hence the product rule tells us that there are $n \cdot n \cdot \dots \cdot n = n^m$ such functions from a set with m elements to one with n elements.

Counting Functions CONT'D

- That is there are 7^4 different functions from a set with four elements to a set with seven elements.

Counting One-to-One Functions

Example 7

How many one-to-one functions are there from a set with m elements to one with n elements?

Solution

- Suppose the elements in the domain are $a_1, a_2, \dots, a_m$.
- There are n ways to choose the value of a_1 and $n - 1$ ways to choose a_2 , etc.
- The product rule tells us that there are $n(n - 1)(n - 2) \dots (n - m + 1)$ such functions.

Product Rule in Terms of Sets

- If $A_1, A_2, \dots, A_m$ are finite sets, then the number of elements in the Cartesian product of these sets is the product of the number of elements of each set.
- This means that the task of choosing an element in the Cartesian product $A_1 \times A_2 \times \dots \times A_m$ is done by choosing an element in A_1 , an element in A_2 , ..., and an element in A_m .

Product Rule in Terms of Sets

- Therefore by the product rule, it follows that:

$$\begin{aligned} n(A_1 \times A_2 \times \dots \times A_m) &\iff |A_1 \times A_2 \times \dots \times A_m| \\ &= |A_1| \cdot |A_2| \cdot \dots \cdot |A_m| \end{aligned}$$

Conclusion

- In this lesson, you have learnt about Product rule and Sum Rule
- If you have any questions about this lecture, please do not hesitate to contact using whatsapp the discussion forum on this platform and we will be happy to assist you.
- I am convinced that this lecture was well presented and you have understood it.
- I wish you all the best.